The Pancreas



Structure, Hormones, Regulation and Disease

Pancreas location and structure

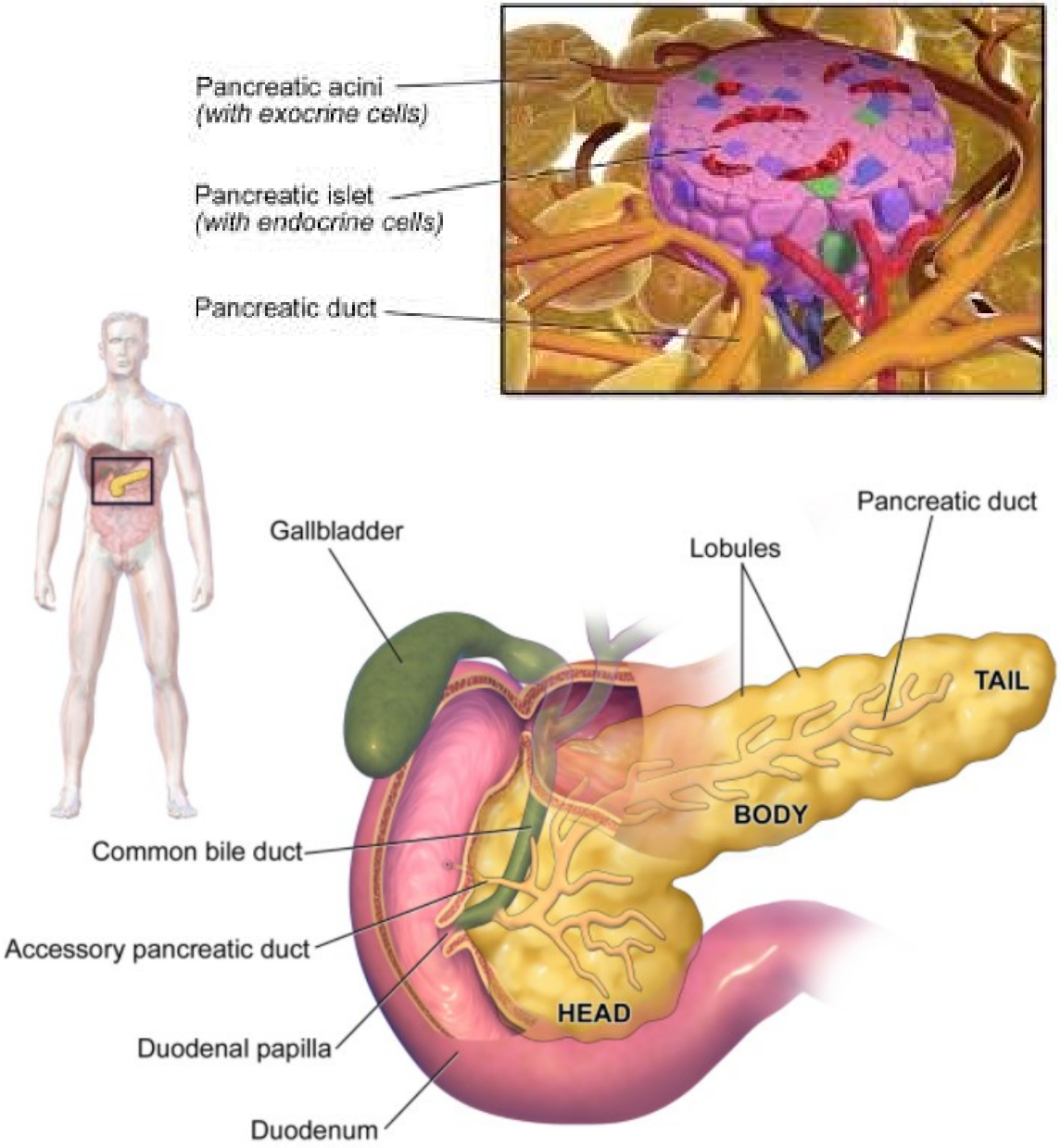
- Located in abdomen, near duodenum
- Leaf shaped organ
- Has both exocrine and endocrine functions

Exocrine:

- Produces amylase and lipase, which travel down ducts to intestine. Involved in digestion

Endocrine:

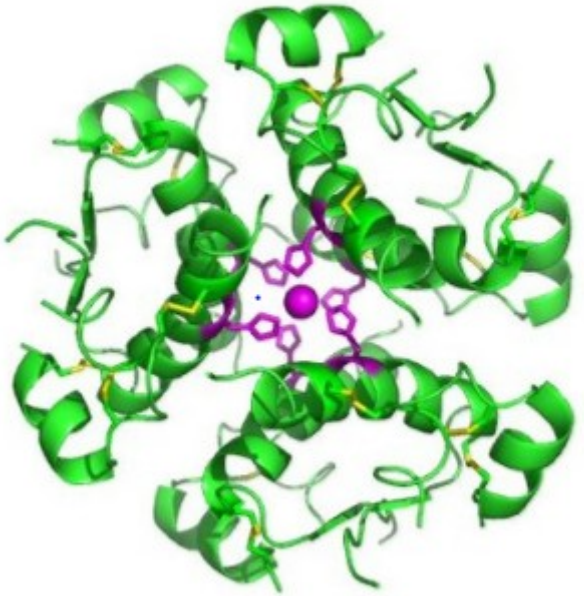
- Produces hormones *insulin* and *glucagon* from clusters of tissue “Islets of Langerhans”
- Regulation of blood sugar levels



Learning Aim: Describe the location and structure of the pancreas.

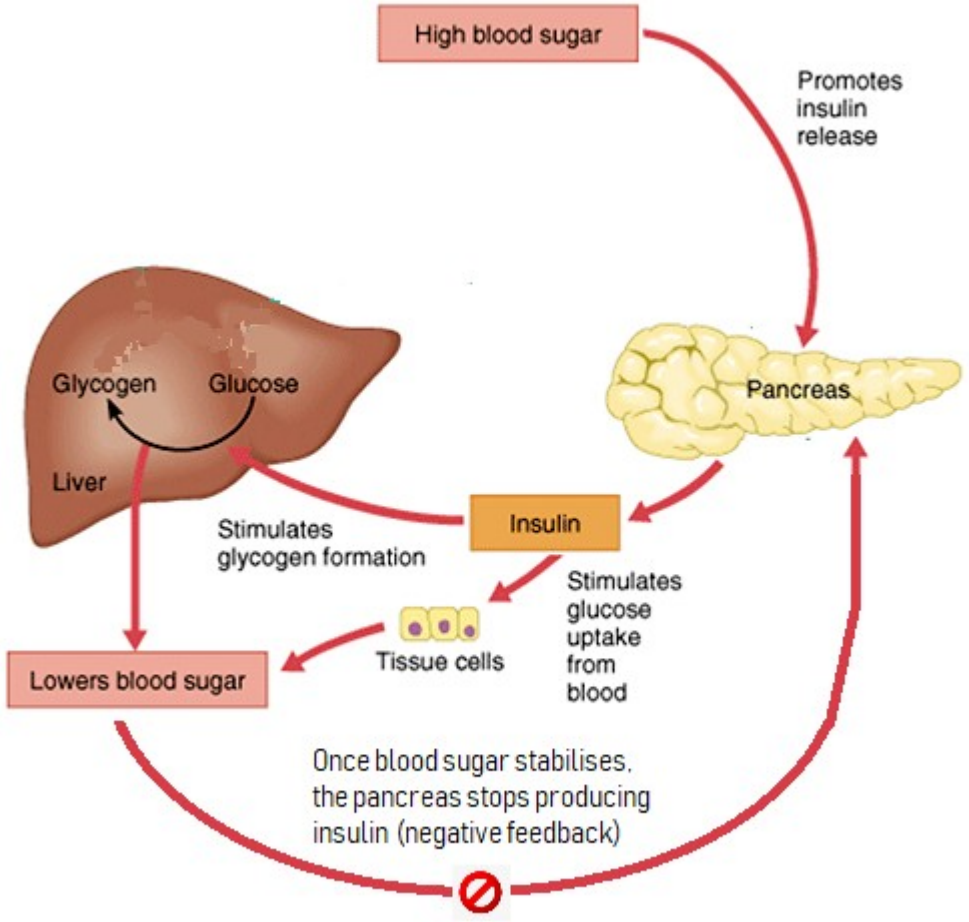
Insulin – production and effects

- Peptide hormone type
- Produced by beta cells of the pancreatic islets.
- Release is stimulated by rising blood glucose levels.
- Binds to receptors on tissue cells, allowing glucose to be absorbed from blood into cells – used for cellular respiration.
- Stimulates conversion of glucose to glycogen in liver and muscle tissue



Learning Aim: Describe the production and effects of insulin.

Insulin regulation



1: High blood sugar (eg because of food intake) is

detected by beta cells of pancreas.

2: The pancreas beta cells release insulin into the bloodstream.

3: Insulin
- stimulates body cells (esp muscle) to take up glucose from the blood.
- Stimulates the liver to convert glucose to ***glycogen*** and store it.

4: As a result, blood sugar levels fall.

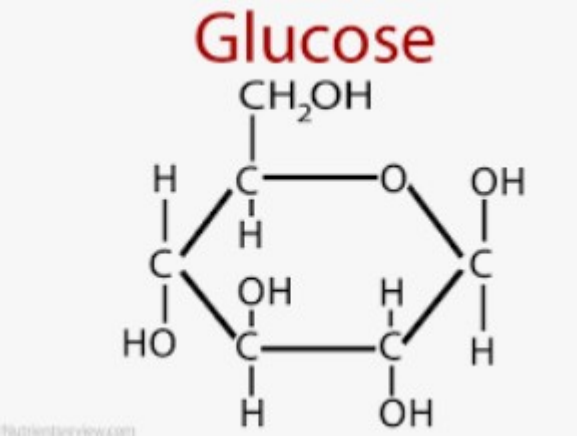
5: Once blood sugar stabilises the pancreas detects this and stops producing insulin (negative feedback).

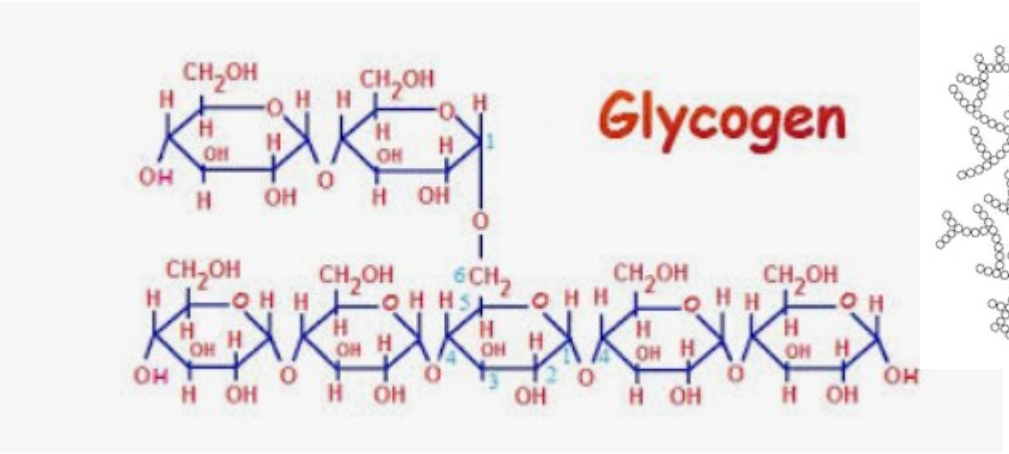
https://youtu.be/OlHez8gwMgw

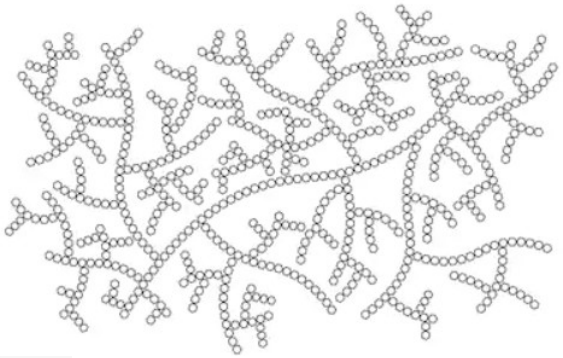
Learning Aim: Explain how blood sugar is regulated via insulin.

Wait, what the heck is glycogen???

- Glucose is stored as glycogen in the liver in response to insulin.
- Glucose is a simple sugar. Glycogen is a more complex sugar:



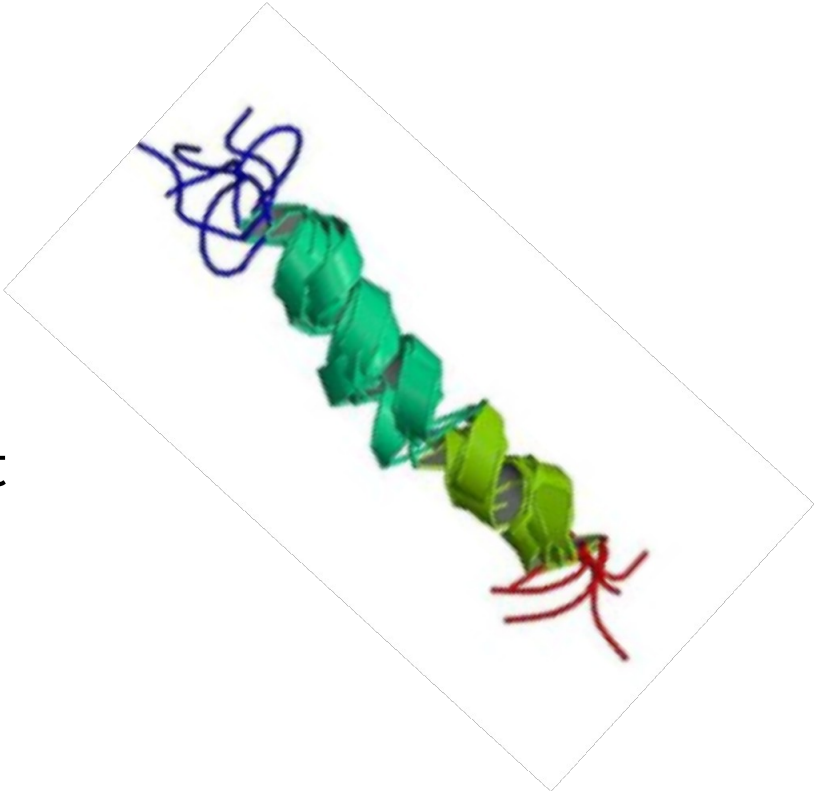




- Glycogen is turned back into glucose when blood sugar is low.

Glucagon – production and effects

- Peptide hormone
- Produced by alpha cells of the pancreatic islets
- Release is stimulated by falling blood glucose levels
- Stimulates liver to convert glycogen to glucose and release it into the blood
- *Note:* ***Glucagon*** *is a hormone.* ***Glycogen*** *is the complex sugar stored in the liver. Glucagon makes the liver break glycogen down into glucose. Don't mix them up! Sorry – it's confusing.*



Learning Aim: Describe the production and effects of glucagon

Glucagon regulation

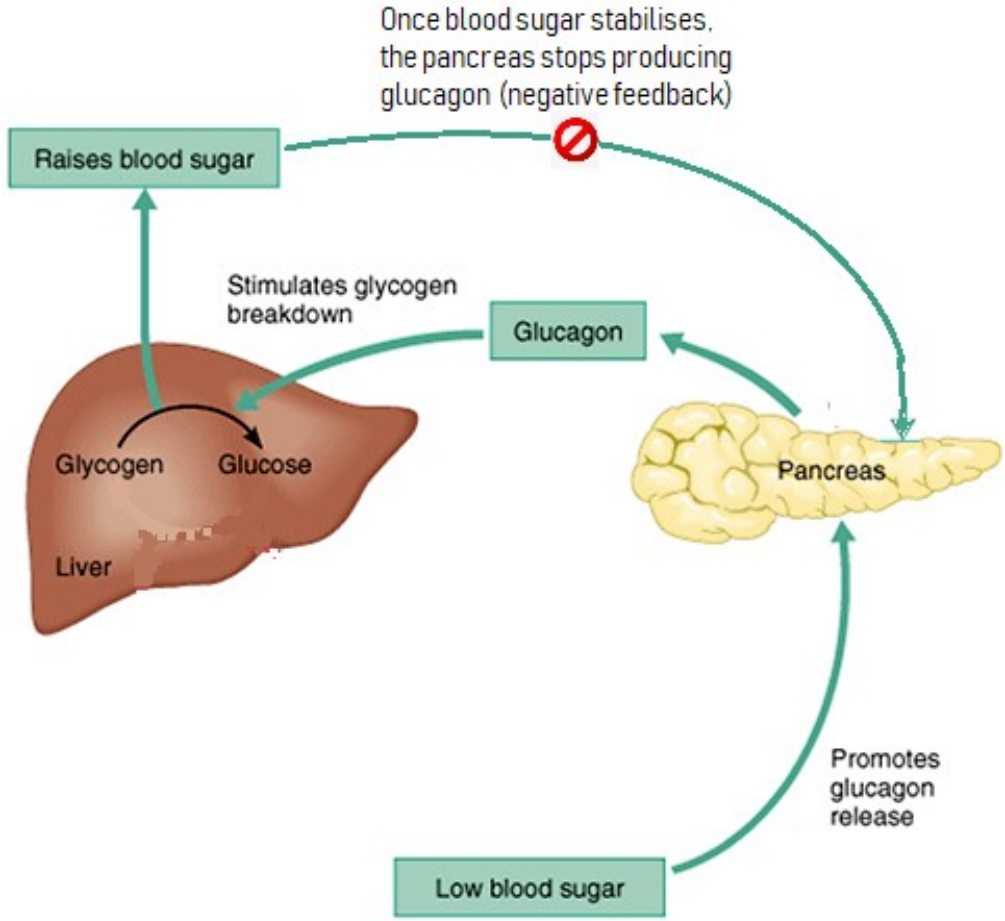
1. Falling blood glucose is detected by alpha cells in the pancreas.

2. The pancreatic alpha cells release glucagon into the blood.

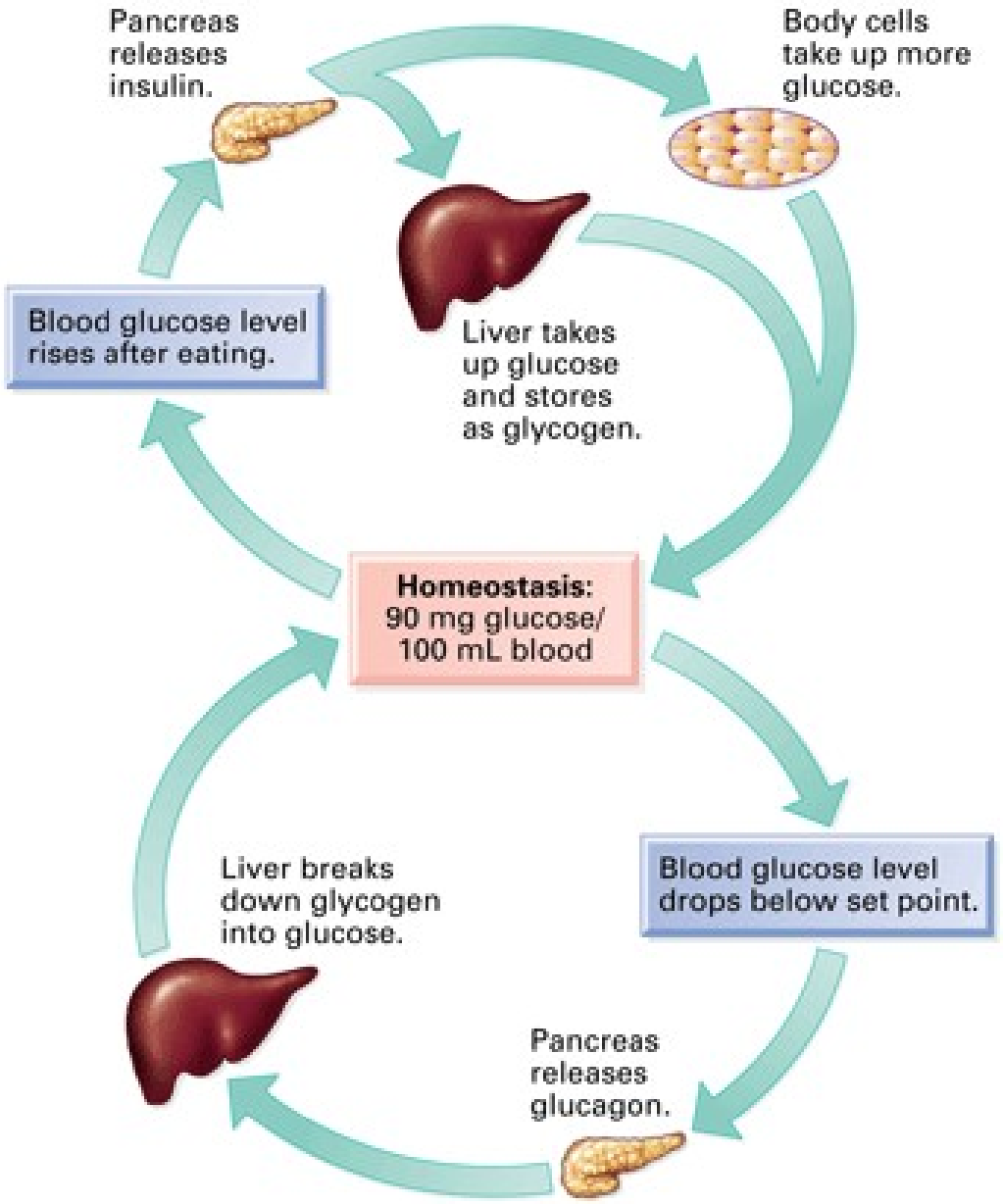
3. Glucagon stimulates the liver to converts glycogen to glucose.

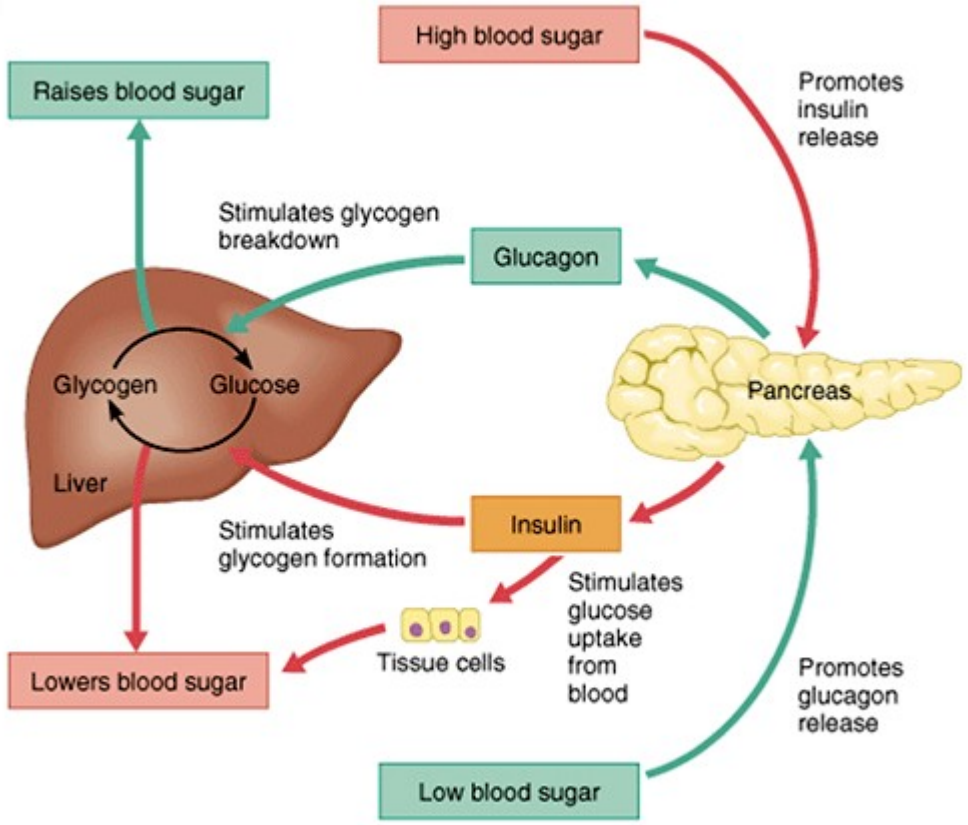
4. Glucose is released into the bloodstream, and blood glucose levels rise

5. Once blood sugar stabilises, the pancreas stops producing glucagon (negative feedback).



Learning Aim: Describe how blood sugar is regulated via glucagon


Pancreas releases insulin.
Body cells take up more glucose.
Blood glucose level rises after eating.
Liver takes up glucose and stores as glycogen.
Homeostasis: 90 mg glucose/ 100 mL blood
Liver breaks down glycogen into glucose.
Blood glucose level drops below set point.
Pancreas releases glucagon.


High blood sugar
Promotes insulin release
Raises blood sugar
Stimulates glycogen breakdown
Glucagon
Glycogen
Glucose
Liver
Pancreas
Insulin
Stimulates glycogen formation
Stimulates glucose uptake from blood
Tissue cells
Lowers blood sugar
Promotes glucagon release
Low blood sugar

Diabetes mellitus: a disorder of glucose regulation

Common Cause(s)	Symptoms	Diagnosis	Treatment
Type 1 Diabetes: • Autoimmune attack of pancreas destroys islet tissue. Little to no insulin produced so glucose cannot be absorbed by body cells.	• Usually childhood onset • Extreme thirst • Excessive urination • Weakness • Fatigue • Diabetic ketoacidosis – odd smelling breath due to ketones • Death if untreated	• Blood test to show high blood glucose levels. • Presence of glucose in urine	Daily insulin injections, accompanied by careful diet and exercise regime.
Type 2 Diabetes: • Lifestyle cause with some genetic components. • Cells become resistant to insulin, so glucose cannot be absorbed	• Adult onset (45+) • Few symptoms at first • As severity increases: • Thirst • Excessive urination • Weakness • Poor wound healing • Headaches • Dizziness		• Lifestyle changes • Weight loss • Regular exercise • Low sugar, good quality diet • Medication to manage insulin resistance

Learning Aim: Describe Diabetes Mellitus Types 1 and 2 in detail